

Using a CSV opening

Define an arbitrary, curved opening profile from a width/height trace instead of the rectangular Space opening.

The CSV format

Two columns, one row per point:

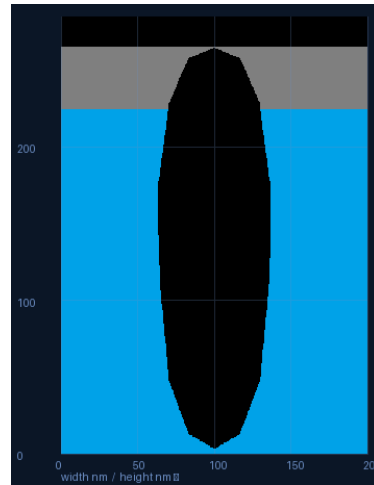
width = the full opening width (CD) at that height — a size, always ≥ 0 .

height = the vertical position (nm) — may be negative (it is a position).

The opening is drawn **symmetric** about the centerline ($\pm \text{width}/2$).

width	height
0	0
34	10
60	45
72	110
74	170
60	225
34	255
0	262

example_opening.csv



The trace loaded as the opening

Steps

1. Prepare a CSV with columns **width,height** (see example). Save it anywhere.
2. In **Base feature**, click “**Load CSV opening...**” and choose your file.
3. **Placement**. By default the trace is **top-aligned** (its top sits at the stack top, cutting downward). To pin it elsewhere, set “**CSV height=0 at (nm)**” — the row where height = 0 lands that many nm from the stack bottom; negative heights sit below that line.
4. While a CSV opening is active, **Space** and **Opening depth** are ignored; **Top vacuum** still applies.
5. If the curved walls look jagged, raise **Smoothing** (Off / 2× / 4× / 8×). Then **Render** and **Save .bmp**.

Tips

- If the material stack is empty, a silicon base is added automatically so the opening is visible.
- Negative width is rejected; ragged rows / blanks are rejected with a clear message.
- Sort isn't required — points are ordered by height internally.